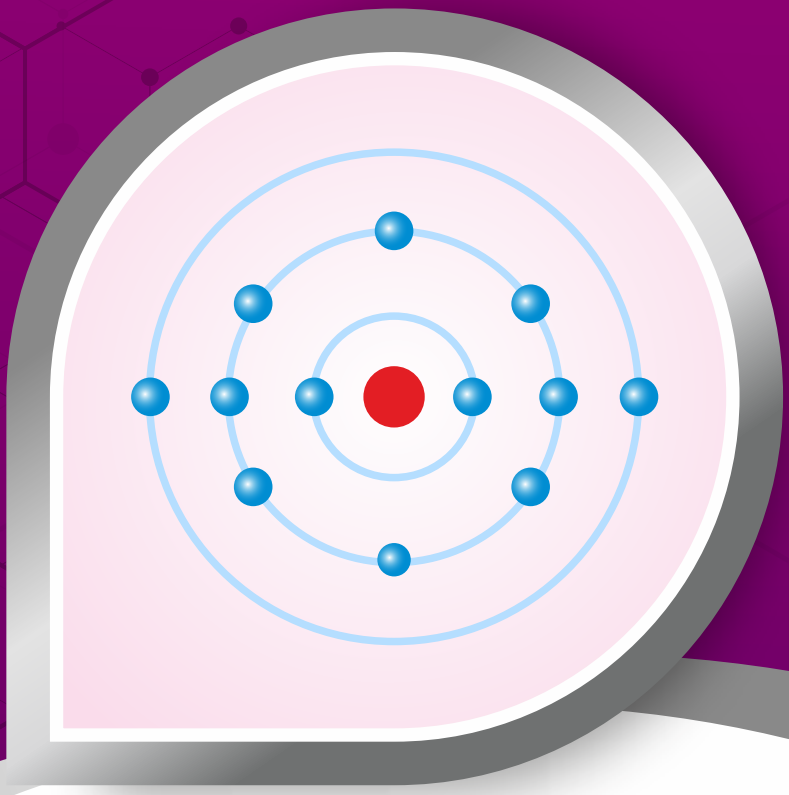
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# INORGANIC CHEMISTRY

## JEE (MAIN+ADVANCED)

NURTURE COURSE



## Exercise

# Quantum Number & Electronic Configuration

(English Medium)

# EXERCISE # O-1

***General Introduction :***

**1.** The total number of neutrons in dipositive zinc ion with mass number 70 is
(A) 34 (B) 40 (C) 36 (D) 38

**QN0001**

**2.** It is known that atom contain protons, neutrons and electrons. If the mass of neutron is assumed to half of its original value where as that of proton is assumed to be twice of its original value then the atomic mass of $^{14}_{6}C$ will be -
(A) same (B) 25% more (C) 14.28 % more (D) 28.5% less

**QN0002**

**3.** Two monoatomic cations $x^{\oplus}$ and $y^{2+}$ are isoelectronic then select the correct statement : (Both elements are consecutive)
(A) Both element x and y have same number of electrons
(B) Total number of valence electrons are more in element x, than element 'y'
(C) Total number of valence electrons are more in element y, than element 'x'
(D) Both (A) and (B) are correct

**QN0003**

**4.** Which of the following sets contain only isoelectronic ions?
(A) $Zn^{2+}$, $Ca^{2+}$, $Ga^{3+}$ , $Al^{3+}$ (B) $K^{+}$, $Ca^{2+}$, $Sc^{3+}$, $Cl^{-}$
(C) $P^{3-}$, $S^{2-}$, $Cl^{-}$, $Zn^{+2}$ (D) $Ti^{4+}$, Ar, $Cr^{3+}$, $V^{5+}$

**QN0004**

***Quantum number***

**5.** Which quantum number will determine the shape of the orbital
(A) Principal quantum number (B) Azimuthal quantum number
(C) Magnetic quantum number (D) Spin quantum number

**QN0005**

**6.** In Palladium (Atomic no.–46), number of electron having ($\ell = 2$) will be -
(A) 20 (B) 18 (C) 16 (D) 22

**QN0006**

**7.** For an electron present in which of the following orbital for which $(n + l + m + s)$ value is maximum. Consider maximum possible value for 'm' and minimum possible value of $m_s$ (where ever applicable).
(A) 3p (B) 5p (C) 4d (D) 5s

**QN0007**

**8.** Choose the correct option for the quantum numbers of the last electron of $K^{+}$.
(A) 4, 0, 0, +1/2 (B) 3, 1, –1, –1/2 (C) 4, 1, 0, $-\frac{1}{2}$ (D) 3, 0, 1, $\frac{1}{2}$

**QN0008**

**9.** Find the sum of maximum number of electrons having +1 and –1 value of 'm' in Ti (Atomic number = 22)
(A) 6 (B) 8 (C) 10 (D) 12

**QN0009**

**10.** The number of electrons in Ca having minimum value of $\left|\frac{n}{\ell \times m_\ell}\right|$ is.
(Consider only non-zero values of $\ell$ and m)
(A) 6 (B) 3 (C) 4 (D) None of these

**QN0010**

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***Electronic Configuration***

**11.** A neutral atom of an element has two K, eight L, nine M and two N electrons then electronic configuration of the element is

(A) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^1$ (B) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$

(C) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3d^2\ 3p^6\ 4s^1$ (D) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$

**QN0011**

**12.** The explanation for the presence of three unpaired electrons in the nitrogen atom can be given by

(A) Pauli's exclusion principle (B) Hund's rule

(C) Aufbau's principle (D) None of the above

**QN0012**

**13.** If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower than that of normal ground state configuration $1s^2\ 2s^2\ 2p^3$ , because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed because it violates :–

(A) Aufbau's principle (B) Hund's rule

(C) Pauli's exclusion principle (D) None of the above

**QN0013**

***Effective Nuclear charge ($Z_{eff}$) :***

**14.** The $Z_{eff}$ for (as Slater's rule)

3d electron of Cr

4s electron of Cr

3d electron of $Cr^{3+}$

3s electron of $Cr^{3+}$ are in the order respectively

(A) 4.6, 2.95, 4.95, 8.05 (B) 4.95, 2.95, 4.6, 8.05

(C) 4.6, 2.95, 5.3, 12.75 (D) none of these

**QN0014**

**15.** Total number of possible shells in uranium atom (atomic no. z = 92)

(A) 7 (B) 1 (C) 6 (D) None of these

**QN0015**

**16.** Which of the following has the maximum number of unpaired electrons ?

(A) $Mg^{2+}$ (B) $Ti^{3+}$ (C) $V^{3+}$ (D) $Fe^{2+}$

**QN0016**

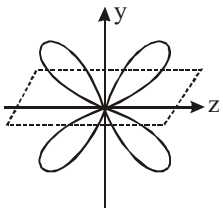
**17.** Gaseous state electronic configuration of nitrogen atom can be represented as :

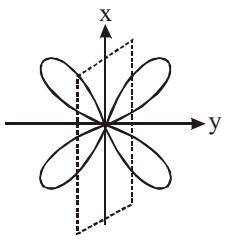
(A) ↑↓ ↑↓ ↑ ↑ ↑ (B) ↑↓ ↑↓ ↑ ↓ ↑

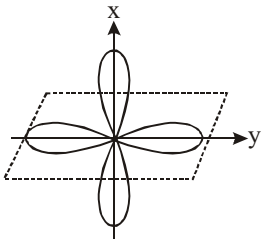
(C) ↑↓ ↑↓ ↑ ↓ ↓ (D) ↑↓ ↑↓ ↓ ↓ ↑

**QN0017**

**18.** Select the nodal plane from the given diagram which is similar to the nodal plane of $p_x$ orbital ?

(A) 


(B) 


(C) 


(D) None of these

**QN0088**

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# EXERCISE # O-2

***General Introduction :***

**1.** Isotones are :
(A) The atoms of different elements
(B) Have same number of neutrons
(C) Have same number of (neutrons + protons)
(D) Have same difference of mass number and atomic number

**QN0018**

***Quantum number***

**2.** For an electron present in which of the following orbital for which (n + *l*) value is maximum.
(A) 3p (B) 5p (C) 4d (D) 5s

**QN0019**

**3.** Correct set of four quantum numbers for valence electron of rubidium( Z = 37) is

(A) 5, 0, 0, $+\frac{1}{2}$ (B) 5, 0, 0, $-\frac{1}{2}$ (C) 5, 1, 1, $+\frac{1}{2}$ (D) 6, 0, 0, $+\frac{1}{2}$

**QN0020**

**4.** The correct set of quantum numbers for the unpaired electron of chlorine atom is

| | **n** | **ℓ** | **m** | | **n** | **ℓ** | **m** |
|---|---|---|---|---|---|---|---|
| (A) | 2 | 1 | 0 | (B) | 2 | 1 | 1 |
| (C) | 3 | 1 | 1 | (D) | 3 | 1 | 0 |

**QN0021**

**5.** Which of the following sets of quantum numbers represent an impossible arrangement ?

| | **n** | **ℓ** | **m** | $\mathbf{m_s}$ | | **n** | **ℓ** | **m** | $\mathbf{m_s}$ |
|---|---|---|---|---|---|---|---|---|---|
| (A) | 3 | 3 | –2 | $\frac{1}{2}$ | (B) | 4 | 0 | 0 | $\frac{1}{2}$ |
| (C) | 3 | 2 | –3 | $\frac{1}{2}$ | (D) | 5 | 3 | 0 | $\frac{1}{2}$ |

**QN0022**

**6.** The quantum numbers for the $19^{th}$ electron of Cr (Z = 24) are
(A) n = 3, $\ell$ = 0, m = 0, $m_s$ = + ½ (B) n = 4, $\ell$ = 0, m = 0, $m_s$ = + ½
(C) n = 3, $\ell$ = 2, m = 2, $m_s$ = + ½ (D) n = 4, $\ell$ = 0, m = 0, $m_s$ = – ½

**QN0023**

**7.** The maximum number of electron having n × $\ell$ × m = 0 in $Zn^{2+}$ is equal to the -
(A) Atomic number of Mg
(B) 12
(C) Total number of electron in Zn which have n + $\ell$ = 0
(D) 'p' electrons in Ar

**QN0024**

***Electronic Configuration***

**8.** The species which have same number of electrons in outer most and penultimate shell -
(A) Ca (B) Ar (C) $V^{+3}$ (D) $Sc^{3+}$

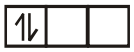
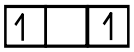
**QN0025**

**9.** Which sub-shell fill completely before the 4f?
(A) 6s (B) 5p (C) 5d (D) 4d

**QN0026**

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**10.** The electronic configuration of a carbon atom is $1s^2$, $2s^2$, $2p^2$ and consider the following four arrangements of the 2p electrons. Which arrangement have lowest energy ?

(A)  (B)  (C)  (D) 

**QN0027**

**11.** Hund's rule is applicable for :-

(A) d-subshell (B) p-subshell (C) s-subshell (D) f-subshell

**QN0028**

**12.** Which of the following has maximum number of unpaired electron.

(A) Fe (B) Fe (II) (C) Fe (III) (D) Mn (II)

**QN0029**

**13.** Mn (Z = 25) = $1s^3\ 1p^9\ 2s^3\ 2p^9\ 2d^1$

Which of the following change is required so that Mn have above ground state electronic configuration :
(Consider only integral values of quantum numbers)

(A) Change in the value of $\ell$ (azimuthal quantum number) for any subshell
(B) Change in the possible values of $\ell$ (azimuthal quantum number)
(C) Change in the Pauli rule
(D) Change in the $(n + \ell)$ rule

**QN0030**

**14.** The number of d- electrons in $Mn^{2+}$ is equal to that of

(A) p-electrons in N (B) s-electron in Na
(C) d-electrons in $Fe^{+3}$ (D) p-electrons in $O^{-2}$

**QN0031**

**15.** Select incorrect statement(s) :

(A) $d_{z^2}$ orbital has different shape from rest of all d-orbitals
(B) For the formation of cation electrons are always removed from 4s.
(C) Zinc is a p-block element.
(D) Principal quantum number depend upon the value of azimuthal quantum number

**QN0032**

**16.** Select the correct statement(s)

(A) In He, maximum degenerate orbitals in 'O' energy level is 9 whereas in $He^+$, it is 25.
(B) In $Li^+$, energy is absorbed/released to change the value of 'n' and '$\ell$' of electron whereas in $Li^{+2}$ energy is absorbed/released only to change the value of 'n' of electron.
(C) In He and Li, energy of 2s subshell is same as both are multielectronic system
(D) Exchange energy also plays a role in differentiating energy of various subshells in same energy level in multielectronic system compared to monoelectronic system.

**QN0089**

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## EXERCISE # S-1

***Integer Answer Type (0 to 9) :***

**1.** Find total no. of orbitals in nickel which have $|m| \leq 1$ and at least one electron is present, where 'm' is magnetic quantum number.

**(Given your ans. as sum of digits for example. If your ans is 57 then 5 + 7 = 12 and 1 + 2 =3)**

**QN0033**

**2.** Minimum number of electrons having $m_s = \left(-\frac{1}{2}\right)$ in Cr is "_____".

**QN0034**

**3.** How many elements are possible for the I$^{st}$ period of periodic table if azimuthal quantum number can have integral values from 0 to (n + 1). [n = shell number & other rules are remaning same to form periodic table.

**QN0035**

**4.** Find number of unpaired electrons when Fe does not follow $(n + \ell)$ rule and filling of electron takes palce shell after shell and Hund's rule is also not obeyed.

**QN0036**

**5.** Find the maximum number of electrons having same $Z_{eff}$ value for sulphur atom

**QN0037**

**6.** Find the sum of maximum unpaired $e^-$ present in one 5g & one 6g orbital.

**QN0038**

**7.** Find out the maximum number of electrons that can involve in the shielding of an electron, having quantum numbers : n = 2, $\ell$ = 1, m = 0, $m_s = +\frac{1}{2}$, in an atom.

**QN0039**

**8.** Find the sum of minimum and maximum possible value of x in $Fe^{+x}$ ion, if magnetic moment of $Fe^{+x}$ = 4.89 B.M.

**QN0040**

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## EXERCISE # S-2

*Paragraph for Question 1 to 3*

The general electronic configuration of outer most and penultimate shell is given as $(n-1)s^2$ $(n-1)p^6$ $(n-1)d^x ns^2$. Then for an element with n = 4 and x = 6.

**1.** The number of protons present in the divalent cation of the element of above configuration is :-
(A) 24 (B) 25 (C) 26 (D) 27

**QN0041**

**2.** The element is :
(A) Mn (B) Fe (C) Co (D) Li

**QN0041**

**3.** The number of unpaired electrons in the divalent cation of the given element in isolated gaseous state is:-
(A) 0 (B) 3 (C) 4 (D) 1

**QN0041**

*Paragraph for Question 4 & 5*

Consider a hypothetical atom where $p_x$, $p_y$, $p_z$, $d_{xy}$, $d_{xz}$, $d_{yz}$ and $d_{x^2-y^2}$ orbitals are present for principal quantum number n = 4.

**4.** Find the number of other orbital which lobes are fully present in the nodal plane of $p_x$ orbital :-
(A) 2 (B) 4 (C) 3 (D) 5

**QN0042**

**5.** Which of the following orbitals lobe is not present at all either in the nodal plane of $p_x$ orbital or in the nodal plane of $p_y$ orbital.
(A) $d_{xy}$ (B) $d_{yz}$ (C) $d_{x^2-y^2}$ (D) $p_z$

**QN0042**

*Paragraph for Question 6 & 7*

Isotopes, Isobars and Iso-diaphers are some basic definitions related to the atom, which are based upon the variation in no. of electron, proton or neutrons.

**6.** Which of the following pair represents the Isobar species :-
(A) $_{18}Ar^{40}$ , $_{6}C^{12}$ (B) $_{8}O^{16}$ , $_{7}N^{14}$ (C) $N_2O$, $CO_2$ (D) None of these

**QN0043**

**7.** Which of the following pair is correct for isosters :-
(A) $N_2O$, $CH_4$ (B) $N_2O$, $CO_2$ (C) $CO_2$, $SO_2$ (D) $N_2O$, HOCl

**QN0043**

**8.**

| **Column-I** **Element** | **Column-II** **Max. value of n and $\ell$ respectively ; consider filled subshell only** |
|---|---|
| (P) P | (1) 6, 2 |
| (Q) N | (2) 3, 1 |
| (R) Pb | (3) 6, 3 |
| (S) Cs | (4) 2, 1 |

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 4 | 1 | 2 | 3 |
| (C) | 3 | 1 | 2 | 4 |

| | P | Q | R | S |
|---|---|---|---|---|
| (B) | 2 | 4 | 1 | 3 |
| (D) | 2 | 4 | 3 | 1 |

**QN0044**

**9.** Match the following:

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| Column–I | Column–II |
|---|---|
| (P) Same number of unpaired electrons are present in (excluding zero) | (1) $Na^+$, $Mg^{2+}$, $F^-$ |
| (Q) Same number of electrons in s & p subshells. | (2) $F^-$, Mg, $O^{2-}$ |
| (R) Same number of electrons with the $l = 1$ | (3) Mg, Ne, $O^{2-}$ |
| (S) Same number of total electrons | (4) Li, Na, K |

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 1, 2 | 3, 4 | 1 | 4 |
| (B) | 4, 2 | 3, 1 | 2, 3 | 4, 1 |
| (C) | 4 | 1 | 1, 2, 3 | 1 |
| (D) | 3 | 1, 2 | 3, 4 | 1, 3, 4 |

**QN0045**

**10.** Match the following :

| Column–I (Orbital) | Column–II (Property) |
|---|---|
| (P) s | (1) Have electron density at all three axes |
| (Q) $p_x$ | (2) YZ plane is nodal plane |
| (R) $d_{xy}$ | (3) dumbell shape |
| (S) $d_{x^2-y^2}$ | (4) have azimuthal quantum no. $\ell = 2$ |

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 1, 3 | 2, 3 | 3 | 3, 4 |
| (B) | 1 | 2, 3 | 2, 4 | 4 |
| (C) | 1 | 2, 3, 4 | 3, 4 | 1, 4 |
| (D) | 2 | 3, 4 | 2, 3 | 1, 4 |

**QN0046**

***Match The Column :***

**11.** Match the following :

| Column–I ($e^-$ configuration) | Column–II (Property) |
|---|---|
| (A) $d^8$ | (P) Symmetrical distribution |
| (B) $d^{10}$ | (Q) Unsymmetrical distribution |
| (C) $d^6$ | (R) No of exchange pair are maximum among these |
| (D) $d^5$ | (S) two electrons must be present in $d_{x^2-y^2}$ orbital |
| | (T) at least one electron is present in orbital having m = –1 |

**QN0047**

**12.** **Column-I** **Column-II**

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(A) $Zn^{2+}$
(B) $Ga^{+}$
(C) $Fe^{3+}$
(D) $Br^{-}$

(P) Diamagnetic
(Q) Spin magnetic moment = $\sqrt{35}$ BM
(R) 18 $e^{-}$ in outer most shell
(S) 3d subshell is fully filled
(T) All the orbital of outer most shell are fully filled

**QN0048**

**MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)**

| Column - I | Column - II | Column - III |
|---|---|---|
| (A) Paramagnetic set | (i) $Na^{+}$,$Mg^{+2}$, $F^{-}$ | (P) same value of principal quantum number for last electron |
| (B) Isoelectronic set | (ii) Li, Na, K | (Q) The non zero number(s) of $e^{-}$(s) for n = 3 and $\ell \geq 1$ is |
| (C) The set for which value(s) of spin multiplicity is $\geq 1$ | (iii) $Fe^{+3}$, $Co^{2+}$, $Ni^{+2}$ | (R) The value of "$m_s$" must be $+\frac{1}{2}$ for last electron |
| (D) The set of elements belongs to same period in periodic table | (iv) $S^{-2}$,$Cl^{-}$, $P^{-3}$ | (S) Set for which the value of m = ±2 is possible for electron(s) |

**13.** Which one of the following options is the **CORRECT** combination?

(A) (A, i, P) (B) (B, iv, S) (C) (D, iii, S) (D) (C, iii, R)

**QN0049**

**14.** Which one of the following options is the **INCORRECT** combination?

(A) (A, iii, P) (B) (C, ii, P) (C) (B, i, P) (D) (B, iv, Q)

**QN0049**

**15.** Which one of the following options is the **CORRECT** set of species with number of nodal planes for filled/partially filled orbitals is $\leq$ 1 for all given species in set?

(A) (B, ii, R) (B) (B, iv, P) (C) (A, i, Q) (D) (D, iii, S)

**QN0049**

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## EXERCISE # JEE-MAIN

**1.** The numbers of protons, electrons and neutrons in a molecule of heavy water are respectively
(1) 10, 10, 10 (2) 8, 10, 11 (3) 10, 11, 10 (4) 11, 10, 10
**[JEE-MAIN, (ONLINE)-2013]**
**QN0053**

**2.** Given **[JEE-MAIN, (ONLINE)-2013]**
(a) n=5, $m_l$= +1 (b) n = 2, $l$ = 1, $m_l$ = –1, $m_s$ = – 1/2
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (a) and (b) are respectively :
(1) 8 and 1 (2) 25 and 1 (3) 2 and 4 (4) 4 and 1
**QN0054**

**3.** The correct set of four quantum numbers for the valence electrons of rubidium atom (Z = 37) is:
**[JEE(Main)-2014]**
(1) $5,1,1,+\frac{1}{2}$ (2) $5,0,1,+\frac{1}{2}$ (3) $5,0,0,+\frac{1}{2}$ (4) $5,1,0,+\frac{1}{2}$
**QN0055**

**4.** If the principal quantum number n = 6, the correct sequence of filling of electrons will be:-
**[JEE-MAIN, (ONLINE)-2015]**
(1) ns→(n–1)d→(n–2)$f$→np (2) ns→np→(n–1)d→(n–2)$f$
(3) ns→(n–2)$f$→(n–1)d→np (4) ns→(n–2)$f$→np→(n–1)d
**QN0056**

**5.** The total number of orbitals associated with the principal quantum number 5 is :
**[JEE-MAIN, (ONLINE)-2016]**
(1) 25 (2) 5 (3) 20 (4) 10
**QN0057**

**6.** The group having isoelectronic species is :- **[JEE-MAIN 2017]**
(1) $O^{2-}$, $F^-$, $Na^+$, $Mg^{2+}$ (2) $O^-$, $F^-$, Na , $Mg^+$
(3) $O^{2-}$, $F^-$, Na , $Mg^{2+}$ (4) $O^-$, $F^-$, $Na^+$, $Mg^{2+}$
**QN0058**

**7.** The isotopes of hydrogen are : **[JEE-MAIN ONLINE 2019]**
(1) Tritium and protium only (2) Deuterium and tritium only
(3) Protium and deuterum only (4) Protium, deuterium and tritium
**QN0059**

**8.** The quantum number of four electrons are given below - **[JEE-MAIN ONLINE 2019]**
I. n = 4, $l$ = 2, $m_l$ = –2, $m_s$ = – ½ II. n = 3, $l$ = 2, $m_l$ = 1, $m_s$ = + ½
III. n = 4, $l$ = 1, $m_l$ = 0, $m_s$ = + ½ IV. n = 3, $l$ = 1, $m_l$ = 1, $m_s$ = – ½
The correct order of their increasing energies will be -
(1) IV < III < II < I (2) IV < II < III < I (3) I < II < III < IV (4) I < III < II < IV
**QN0060**

**9.** The isoelectronic set of ions is : **[JEE-MAIN ONLINE 2019]**
(1) $N^{3-}$, $Li^+$, $Mg^{2+}$ and $O^{2-}$ (2) $Li^+$, $Na^+$, $O^{2-}$ and $F^-$
(3) $F^-$, $Li^+$, $Na^+$ and $Mg^{2+}$ (4) $N^{3-}$, $O^{2-}$, $F^-$ and $Na^+$
**QN0061**

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**10.** The number of orbitals associated with quantum numbers n = 5, $m_s = +\frac{1}{2}$ is : **[JEE-MAIN ONLINE 2020]**

(1) 11 (2) 25 (3) 15 (4) 50

**QN0062**

**11.** Hydrogen has three isotopes (A), (B) and (C). If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the sum of (x), (y) an (z) is : **[JEE-MAIN ONLINE 2020]**

(1) 4 (2) 3 (3) 2 (4) 1

**QN0063**

**12.** The number of orbitals with n = 5, $m_1 = +2$ is _____. (Round off to the Nearest Integer). **[JEE MAINS 2021]**

**QN0086**

**13.** The correct decreasing order of energy, for the orbitals having, following set of quantum numbers: **[JEE MAINS 2022]**

(A) n = 3, $\ell = 0$, m = 0

(B) n = 4, $\ell = 0$, m = 0

(C) n = 3, $\ell = 1$, m = 0

(D) n = 3, $\ell = 2$, m = 1

(A) (D) > (B) > (C) > (A) (B) (B) > (D) > (C) > (A)

(C) (C) > (B) > (D) > (A) (D) (B) > (C) > (D) > (A) **QN0090**

**14.** Which of the following sets of quantum numbers is not allowed ? **[JEE MAINS 2022]**

(1) n = 3, $\ell = 2$, $m_\ell = 0$, s = +1/2

(2) n = 3, $\ell = 2$, $m_\ell = -2$, s = +1/2

(3) n = 3, $\ell = 3$, $m_\ell = -3$, s = –1/2

(4) n = 3, $\ell = 0$, $m_\ell = 0$, s = –1/2 **QN0091**

**15.** Consider the following statements : **[JEE MAINS 2022]**

(A) The principal quantum number 'n' is a positive integer with values of 'n' = 1, 2, 3, ….

(B) The azimuthal quantum number '*l*' for a given 'n' (principal quantum number) can have values as '*l*' = 0, 1, 2, …. n

(C) Magnetic orbital quantum number '$m_l$' for a particular '*l*' (azimuthal quantum number) has (2*l* + 1) values.

(D) ±1/2 are the two possible orientations of electron spin.

(E) For *l* = 5, there will be a total of 9 orbital.

Which of the above statements are correct?

(A) (A), (B) and (C)

(B) (A), (C), (D) and (E)

(C) (A), (C) and (D)

(D) (A), (B), (C) and (D)

**QN0092**

**16.** The electronic configuration of Pt (atomic number 78) is: **JEE MAINS 2022]**

(A) [Xe] $4f^{14}$ $5d^9$ $6s^1$ (B) [Kr] $4f^{14}$ $5d^{10}$

(C) [Xe] $4f^{14}$ $5d^{10}$ (D) [Xe] $4f^{14}$ $5d^8$ $6s^2$ **QN0093**

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## EXERCISE # JEE-ADVANCED

**1.** The maximum number of electrons that can have principal quantum number, n = 3, and spin quantum number, $m_s = -1/2$, is **[JEE 2011]**

**QN0064**

**2.** In an atom, the total number of electrons having quantum numbers n=4, $|m_\ell| = 1$ and $m_s = -\frac{1}{2}$ is: **[JEE Advanced 2014]**

**QN0065**

**3.** Not considering the electronic spin the degeneracy of the second excited state (n = 3) of H-atom is 9, where the degeneracy of the second excited state of $H^-$ is **[JEE Advanced 2015]**

**QN0066**

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# ANSEWR KEY

## EXERCISE # O-1

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | C | C | B | B | A | C | B | C | C |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | | |
| Ans. | B | B | C | C | D | D | A | D | | |

## EXERCISE # O-2

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A, B, D | B, C | A, B | C, D | A, C | B, D | A, B, D | B, D | A, B, D | C, D |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | | | | |
| Ans. | A, B, D | C, D | B,C,D | B, C | B, C, D | A,B | | | | |

## EXERCISE # S-1

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Ans. | 4 | 9 | 8 | 0 | 8 | 2 | 9 | 6 |

## EXERCISE # S-2

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | C | B | C | C | A | D | B | D | C | B |
| Que. | 11 | | | | | 12 | | | | |
| Ans. | (A)-Q,T (B)-P,R,S,T (C)-Q,T (D)-P,T | | | | | (A)-P,R,S,T (B)-P,S (C)-Q (D)-P,S | | | | |
| Que. | 13 | 14 | 15 | | | | | | | |
| Ans. | C | B | B | | | | | | | |

## EXERCISE # JEE-MAIN

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 1 | 1 | 3 | 3 | 1 | 1 | 4 | 2 | 4 | 2 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | | | | |
| Ans. | 2 | 3 | A | 3 | C | A | | | | |

## EXERCISE # JEE-ADVANCED

| Que. | 1 | 2 | 3 |
|---|---|---|---|
| Ans. | 9 | 6 | 3 |

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